

Easy Pivot Workbench --- Installation Guide

Easy Pivot Workbench is a PHP web application. It can be deployed on a Windows computer using PHP's built-in web server, or deployed to an existing PHP-enabled web server.

Step 1 --- Download and Extract Easy Pivot

Easy Pivot Workbench is distributed through GitHub.

Easy Pivot project:\

https://github.com/pivot-my-stuff/easy_pivot

If you are not using Git, you can download Easy Pivot as a ZIP file.

1. Open the Easy Pivot project using the link above.
2. Click the green Code button.
3. Select Download ZIP.
4. Save the ZIP file to your computer.
5. Extract the ZIP file.

After extracting the ZIP file, open the extracted folder.

You should see a folder named:

`easy_pivot_workbench`

Open it.

Inside, you should see another `easy_pivot_workbench` folder.

That is the Easy Pivot Workbench folder you want to use.

*****Using Git?**** If you already use Git, you can clone or pull the Easy Pivot repository instead. Git users can proceed once the Easy Pivot files are available locally.*

Step 2 --- Install the Easy Pivot Database Stored Procedure

Before using Easy Pivot Workbench, install the Easy Pivot stored procedure for the database platform you intend to use.

The Easy Pivot project contains a database folder for each supported database platform:

```
easy_pivot_workbench
  ■■■■ mysql
  ■■■■ oracle
  ■■■■ postgresql
  ■■■■ sql_server
```

Open the folder for your database platform and locate the supplied Easy Pivot stored-procedure script.

Install the procedure in the database where you want Easy Pivot to be available.

The procedure is installed at the database level. You do not need to create a special database for Easy Pivot, and you do not need to install Easy Pivot across the entire database server.

For example, a database administrator can choose to make Easy Pivot available in one database while leaving it unavailable in another.

The Easy Pivot Workbench database connection must point to a database where the Easy Pivot procedure has been installed.

Database-specific installation

- **MySQL** --- install the procedure from the mysql folder.
- **Oracle** --- install the procedure from the oracle folder.
- **PostgreSQL** --- install the procedure from the postgresql folder.
- **SQL Server** --- install the procedure from the sql_server folder.

The exact installation method depends on the database platform. Use the appropriate database tool to execute the supplied script.

***Important:** Installing the stored procedure and configuring a Workbench database connection are two separate steps. The procedure must exist in the target database before Easy Pivot can use that database.*

Step 2.1 --- Understanding Easy Pivot Connections

Easy Pivot Workbench and the Easy Pivot stored procedure are separate

components.

The Workbench is the application interface. The stored procedure is installed inside the target database.

A Workbench installation can connect to a database located on another computer, provided the database server accepts the connection and the required network access, credentials, and permissions are available.

The connection identifies the database target using the host, port, and database fields:

```
Host      = computer running the database server
Port      = database network port
Database  = database containing the Easy Pivot procedure
```

For example:

```
CRYPTID:3306/baan
```

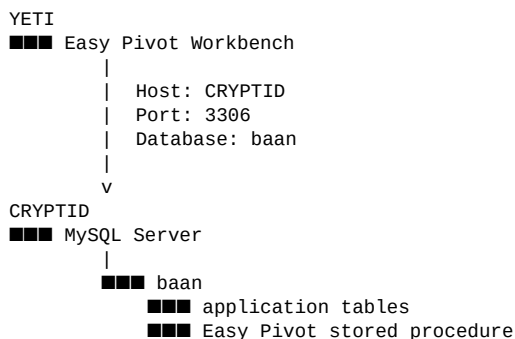
means:

```
Server:   CRYPTID
Port:     3306
Database: baan
```

The database does not have to be named easy_pivot.

Easy Pivot can be installed into any database where the administrator wants to use it. The Workbench connection simply needs to point to that database.

For example, a Workbench hosted on YETI can connect to a MySQL database on CRYPTID:



The Workbench itself does not need to be installed on the same computer as the database server.

Connection information is deployment-specific. A stand-alone Workbench and a Workbench hosted on a web server may connect to the same database, but each deployment must have its own connection configured.

Connection testing

The Test Connection function checks more than network connectivity.

For MySQL, the Workbench reports:

1. Whether the database connection succeeded.
2. The MySQL server version.
3. Whether the Easy Pivot stored procedure exists in the selected database.

A successful test looks like:

```
Connection successful.  
MySQL version: 8.0.46  
Easy Pivot procedure found.
```

If the database connection succeeds but the procedure has not been installed in that database, the test reports:

```
Connection successful.  
MySQL version: 8.0.46  
Easy Pivot procedure NOT found.
```

This distinction is useful when troubleshooting deployment. A successful connection with a missing procedure means that the network connection, database server, and credentials are working; the remaining step is to install the Easy Pivot procedure in the selected database.

*****Important:**** The stored procedure must exist in the database named by the Workbench connection. Installing the procedure in one database does not make it available automatically in other databases on the same server.*

Step 3 --- Install and Verify PHP on Windows

What the stand-alone deployment requires

The Easy Pivot Workbench stand-alone deployment requires PHP 8.x for Windows, with PDO and the PDO driver for the database you intend to use available to PHP.

It does not require Apache, IIS, XAMPP, or another web server. The stand-alone launcher uses PHP's built-in development web server.

PHP is a prerequisite for Easy Pivot and is not included with the Easy Pivot repository.

Required PHP database support

Easy Pivot uses PHP Data Objects (PDO) to communicate with the database.

PDO itself is normally enabled by default in the official Windows PHP distribution, but it should be verified. The database-specific PDO driver must also be available.

For example:

- **MySQL** --- PDO and pdo_mysql
- **PostgreSQL** --- PDO and pdo_pgsql
- **SQL Server** --- PDO and pdo_sqlsrv
- **Oracle** --- PDO and pdo_oci

If either PDO or the required database driver is missing, Easy Pivot will not be able to connect to the database.

***Important:** PHP installation is not complete for Easy Pivot until both PDO and the appropriate database-specific PDO driver have been verified.*

3.1 Check whether PHP is already installed

Open a new Windows Command Prompt and run:

```
php -v
```

If PHP is installed and available to Windows, you should see something similar to:

```
PHP 8.x.x (cli) ...
```

Then run:

```
where php
```

You should see the location of php.exe, for example:

```
C:\php\php.exe
```

If both commands work, PHP is ready for Easy Pivot.

3.2 If PHP is not installed

Download PHP from the official PHP for Windows project:

<https://windows.php.net/download/>

For a normal modern 64-bit Windows computer, choose the current x64 PHP build.

For the stand-alone Easy Pivot Workbench, use the Non Thread Safe (NTS) ZIP distribution.

You do not need the Debug Pack, Development package, Apache, IIS, or XAMPP.

3.3 Extract PHP

Create a directory such as:

```
C:\php
```

Extract the downloaded PHP ZIP file into that directory. When finished, you should have:

```
C:\php\php.exe
```

You can test the installation directly:

```
C:\php\php.exe -v
```

3.4 Add PHP to the Windows PATH

Easy Pivot's launcher executes:

```
php -S localhost:18743
```

Windows therefore needs to know where PHP is located.

1. Open Start and search for environment variables.
2. Select Edit the system environment variables.
3. Click Environment Variables....
4. Under User variables, select Path and click Edit.
5. Click New and enter C:\php.
6. Click OK until all dialogs are closed.

Do not replace the existing PATH. Add the PHP directory as a new entry.

3.5 Verify PHP and database support

Close any Command Prompt windows that were already open before changing PATH.

Open a new Command Prompt and run:

```
php -v  
where php
```

Then verify PDO and the driver for your database.

For MySQL, for example:

```
php -m | findstr /I "PDO mysql"
```

You should see:

```
PDO  
pdo_mysql
```

For other databases, verify the corresponding driver:

```
PostgreSQL → pdo_pgsql  
SQL Server → pdo_sqlsrv  
Oracle     → pdo_oci
```

If PDO or the required database driver is not listed, the PHP installation must be configured before Easy Pivot can connect to that database.

After changing php.ini, restart PHP/the web server as appropriate and repeat the verification.

If php -v and where php work and the required PDO support is present, PHP is ready for Easy Pivot.

Step 4 --- Put Easy Pivot on the Desktop and Create a Shortcut

Open the Easy Pivot folder you identified in Step 1.

Copy the easy_pivot_workbench folder and paste it onto your Windows Desktop.

Open the easy_pivot_workbench folder on your Desktop and locate:

```
start_easy_pivot_workbench.bat
```

Right-click the .bat file and select:

Show more options → Send to → Desktop (create shortcut)

Return to your Desktop and rename the new shortcut:

Easy Pivot Workbench

You can now double-click the Easy Pivot Workbench shortcut to launch

the stand-alone Workbench.

Keep the `easy_pivot_workbench` folder on your Desktop. The launcher needs the files and folders contained within it.

Step 5 --- Start Easy Pivot Workbench

Double-click the Easy Pivot Workbench desktop shortcut.

The launcher checks port 18743, starts PHP's built-in web server, and opens Easy Pivot in your browser.

Port used by the stand-alone launcher

The stand-alone Workbench uses port 18743 by default.

Before starting PHP, the launcher checks whether that port is already in use.

If you previously started Easy Pivot Workbench, the PHP server may still be running even if you closed the browser. Look for the Easy Pivot Workbench command prompt and close it before starting Easy Pivot again.

If Easy Pivot is not already running, another application may be using port 18743. In that case, the `PORT` value in the launcher can be changed.

The Easy Pivot Workbench command prompt

The command prompt window that appears is the PHP server.

Do not close that command prompt while using Easy Pivot.

Closing the browser does not stop the PHP server.

Bookmark Easy Pivot Workbench

When the browser opens Easy Pivot Workbench, we recommend adding the page to your browser's bookmarks or favorites.

The stand-alone Workbench normally opens at:

`http://localhost:18743`

If you close the browser while the Easy Pivot PHP server is still running, you can reopen the browser and select the Easy Pivot Workbench bookmark. The browser will reconnect to the running PHP server without requiring you to start Easy Pivot again.

Stopping the stand-alone Workbench

When you are finished with Easy Pivot, close the Easy Pivot Workbench command-prompt window. This stops the local PHP server.

If you later launch Easy Pivot again while the previous PHP server is still running, the launcher will detect that port 18743 is already in use and tell you how to proceed.

Step 6 --- Deploy Easy Pivot to a Web Server

Easy Pivot Workbench is a PHP web application. The same Workbench used by the stand-alone deployment can also be deployed to an existing web server.

The web server must:

- Support PHP.
- Be configured to execute PHP files.
- Have the Easy Pivot files available to the web server.

Easy Pivot does not install or configure the web server.

Apache example

If your web server is Apache, copy the Easy Pivot Workbench files into Apache's web document directory, normally the htdocs folder.

For example:

```
htdocs
├── easy_pivot_workbench
│   ├── ...
│   └── ...
```

The Workbench can then be accessed through the URL assigned to that directory, for example:

```
http://your-server/easy_pivot_workbench/
```

The exact location of the htdocs directory and the URL depend on the

Apache installation and configuration.

Other web servers

Easy Pivot is not limited to Apache.

If your organization uses IIS or another PHP-enabled web server, follow that server's normal procedure for deploying PHP applications.

Your web-server administrator is responsible for configuring the server and PHP.

Remote database connections

The web server and database server do not have to be the same computer.

The Workbench uses the connection's host, port, and database values to identify the database target. For a remote database, the database server must be configured to accept connections from the web server, and any required firewall rules, authentication, and database permissions must allow the connection.

For example:

```
Easy Pivot Workbench on YETI
    |
    |  CRYPTID:3306/baan
    v
MySQL on CRYPTID
```

This allows one Workbench installation to be used with a database located elsewhere on the network.

PHP database extensions

The PHP installation used by the web server must have PDO and the PDO driver required by the selected database platform available and enabled.

PDO itself is normally enabled by default in the official Windows PHP distribution, but the database-specific driver must be available. The same database-driver requirement applies to the stand-alone and web deployments.

For example:

- **MySQL** --- pdo_mysql
- **PostgreSQL** --- pdo_pgsql

- **SQL Server** --- pdo_sqlsrv
- **Oracle** --- pdo_oci

If the required PHP database driver is not enabled, Easy Pivot will not be able to establish the database connection and may report:

```
Connection failed.
could not find driver
```

This is a PHP/server configuration issue, not an Easy Pivot database connection setting.

Your web-server administrator should verify that the appropriate PHP database extension is installed and enabled before troubleshooting the Easy Pivot connection itself.

Step 7 --- Stand-Alone or Web Deployment?

Easy Pivot Workbench is the same web application in both deployment models. The difference is where PHP hosts it.

Stand-alone deployment	Web deployment
------------------------	----------------

Easy Pivot Workbench Same Same

PHP User installs Server administrator
provides

Web server PHP built-in server Existing PHP-enabled
server

Launcher Included BAT file Not required

Typical access localhost:18743 Server URL

Intended use Individual Windows Shared/server
computer environment

Stand-alone deployment

Use this when you want Easy Pivot on your own Windows computer.

You need:

- Windows
- PHP
- Easy Pivot Workbench
- A web browser

The supplied launcher starts PHP's built-in web server for you.

Web deployment

Use this when Easy Pivot is going to live on an existing web server and be accessed through a normal web URL.

You need:

- An existing PHP-enabled web server
- Easy Pivot Workbench files
- A web browser

Your web-server administrator handles the server and PHP configuration.

Using both deployments

You can use the stand-alone and web deployments of Easy Pivot Workbench at the same time.

For example, you can have the stand-alone Workbench running on your Windows computer while also accessing an Easy Pivot Workbench installation hosted on a web server.

The two deployments operate independently. However, database connections must be configured separately for each deployment because the connection information is stored differently in the stand-alone and web environments.

Configuring a database connection in the stand-alone Workbench does not automatically make that connection available to the web deployment, and vice versa.

The same database can be accessed by both deployments, provided each deployment has been configured with the appropriate connection information and the necessary database access is available.

Which should I use?

If you are installing Easy Pivot for yourself on a Windows computer, use the stand-alone deployment.

If your organization already has a PHP-enabled web server and wants Easy Pivot hosted there, use the web deployment.

Easy Pivot Workbench documentation --- beta